

REQUIREMENT

=====

Build an employee leave application for our organization.

ARCHITECTURE

=====

4. ARCHITECTURE DESIGN

Architecture and Diagram Summary

- **Frontend Layer:** A web-based interface for employees to submit leave requests and view their leave status.
- **API Layer:** RESTful APIs to handle requests from the frontend and communicate with the application services.
- **Application Services Layer:** Core business logic for processing leave requests, approvals, and balance tracking.
- **Persistence Layer:** Database schema and data access objects to store and retrieve leave data.
- **Security Layer:** Authentication and authorization mechanisms to ensure only authorized users can access the system.
- **External Systems:** Integration with existing HR systems for user data and leave policy enforcement.

Code Design

- `com.example.leave.web`: Controllers and request mappings for the API.
- `com.example.leave.service`: Service classes implementing business logic.
- `com.example.leave.repository`: JPA repositories for database access.
- `com.example.leave.model`: Domain objects representing leave requests and employees.
- `com.example.leave.security`: Security configurations and user details service.
- `com.example.leave.validation`: Custom validation annotations and validators.

Implementation Plan

Phase 1: Setup

Tasks:

- Initialize the project with Spring Boot
- Configure PostgreSQL database
- Set up Spring Security

Dependencies: None

Expected Outcome: A running Spring Boot application with a configured database and basic security setup.

Phase 2: Database

Tasks:

- Design and create database schema
- Implement JPA entities and repositories

Dependencies: Phase 1

Expected Outcome: A database schema ready for leave data storage and retrieval.

Phase 3: Domain

Tasks:

- Define domain models for leave requests and employees
- Implement domain logic for leave balance calculations

Dependencies: Phase 2

Expected Outcome: Domain models and logic for managing leave requests and employee data.

Phase 4: Business Logic

Tasks:

- Implement service layer for leave request processing
- Define leave approval workflow

Dependencies: Phase 3

Expected Outcome: Complete business logic for leave request submission and approval.

Phase 5: APIs

Tasks:

- Develop RESTful APIs for leave requests
- Implement API validation and error handling

Dependencies: Phase 4

Expected Outcome: Functional APIs for leave request submission and status checking.

Phase 6: Security and Deployment

Tasks:

- Configure role-based access control
- Prepare application for deployment

Dependencies: Phase 5

Expected Outcome: A secure application ready for production deployment.

API DESIGN

=====

5. API DESIGN

Endpoints

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s S t a t u s	E r r o r S t a t u s
P O S T	/leave-requests	S u b m i t a n e w l e a v e r e q u e s t	{"employeeId":1,"leaveType":"SICK","startDate":"2023-10-01","endDate":"2023-10-05"}	{"id":1,"status":"PENDING"}	201	404500

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r S t a t u s
GET	/leave-requests/{id}	Retrieve leave request by ID	N/A	{"id":1,"employeeId":1,"leaveType":"SICK","startDate":"2023-10-01","endDate":"2023-10-05","status":"PENDING"}	200	404

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r s t a t u s
PUT DELETE	/leave-requests/{id}/approve	Approve a leave request	N/A	{"id":1,"status":"APPROVED"}	200	404 404 403 500

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r s t a t u s
P A T C H	/leave- requests/{id}/ reject	R e j e c t a l l e a v e r e q u e s t	N/A	{"id":1,"status":"REJECTED"}	2 0 0 4 0 4 4 0 3 5 0 0	4 0 0 4 0 4 0 3 5 0 0

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r S t a t u s
GET	/leave-balance/{employeeId}	Get leave balance for an employee	N/A	{"employeeId":1,"leaveType":"SICK","balance":10}	200	404

Validation and Error Handling

Field Level Validation Rules

- employeeId: Required, must be a valid employee ID.

- leaveType: Required, must be one of ["SICK", "VACATION", "PERSONAL"].
- startDate: Required, must be a valid date, must not be in the past.
- endDate: Required, must be a valid date, must be after startDate.
- status: Must be one of ["PENDING", "APPROVED", "REJECTED"].

Standard Error Response Envelope

```
{
  "timestamp": "2023-10-01T12:34:56.789Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid leave request.",
  "details": "The leave request contains invalid data."
}
```

Business Failures to HTTP Status Codes

Business Failure	HTTP Status Code
Invalid request data	400
Leave request not found	404
Unauthorized access	403
Internal server error	500

DATA MODEL

=====

6. DATA MODEL

Model

Table	Column	Data Type	Nullabl e	Ke y	Default	Descriptio n
Employee	id	UUID	No	PK	-	Unique identifier for the employee
Employee	name	VARCHAR	No	-	-	Employee's full name
Employee	email	VARCHAR	No	-	-	Employee's email address

Table	Column	Data Type	Nullabl e	Ke y	Default	Descriptio n
LeaveReques t	id	UUID	No	PK	-	Unique identifier for the leave request
LeaveReques t	employee_id	UUID	No	FK	-	Employee who requested the leave
LeaveReques t	start_date	DATE	No	-	-	Start date of the leave
LeaveReques t	end_date	DATE	No	-	-	End date of the leave
LeaveReques t	status	VARCHAR	No	-	-	Status of the leave request
LeaveReques t	created_at	TIMESTAM P	No	-	CURRENT_TIMESTAM P	Creation timestamp
LeaveReques t	updated_at	TIMESTAM P	No	-	CURRENT_TIMESTAM P	Last update timestamp

ER Diagram

```

{
  "entities": [
    { "name": "Employee", "fields": ["id", "name", "email"] },
    { "name": "LeaveRequest", "fields": ["id", "employee_id", "start_date", "end_date", "status"] },
  ],
  "relationships": [
    { "from": "Employee", "to": "LeaveRequest", "label": "1:N" }
  ]
}

```

Schema Script

```

CREATE TABLE Employee (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR NOT NULL,
    email VARCHAR NOT NULL UNIQUE
);

CREATE TABLE LeaveRequest (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    employee_id UUID NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    status VARCHAR NOT NULL CHECK (status IN ('Pending', 'Approved', 'Rejected')),
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (employee_id) REFERENCES Employee(id) ON DELETE CASCADE
);

CREATE INDEX idx_leaverequest_employee_id ON LeaveRequest(employee_id);
CREATE INDEX idx_leaverequest_status ON LeaveRequest(status);

```

Constraints and Indexes

- **Primary Keys:** id in Employee and LeaveRequest.
- **Foreign Keys:** employee_id in LeaveRequest referencing id in Employee.
- **Unique Constraints:** email in Employee.
- **Check Constraints:** status in LeaveRequest to ensure it is one of 'Pending', 'Approved', or 'Rejected'.
- **Indexes:**
 - idx_leaverequest_employee_id: To speed up queries that filter or join on employee_id.
 - idx_leaverequest_status: To speed up queries that filter on the status of leave requests.

DOCUMENTATION WITH REMAINING PLACEHOLDERS

=====

User Guide

Overview

The Employee Leave Application is designed for employees within our organization to submit leave requests and view their leave status. This application streamlines the leave management process, ensuring that employees can easily request time off and managers can efficiently approve or reject these requests. It also helps in tracking leave balances and maintaining accurate records of leave activities.

Getting Started

To start using the Employee Leave Application, follow these steps:

1. **Login:** Access the application via the web interface and log in using your employee credentials.
2. **Submit a Leave Request:** Navigate to the 'Leave Requests' section and click on 'Submit Leave Request'. Fill in the required details such as leave type, start date, and end date.
3. **Verify Submission:** After submitting the request, you should see a confirmation message with the leave request ID and status as 'PENDING'.

User Roles and Permissions

Employee

- **Capabilities:** Submit leave requests, view their own leave status, and check their leave balance.
- **Typical workflows:** Submit a leave request, check the status of a leave request, view leave balance.
- **Restrictions:** Cannot approve or reject leave requests, cannot view other employees' leave status.

Manager

- **Capabilities:** Approve or reject leave requests submitted by employees under their supervision.
- **Typical workflows:** Review a leave request, approve or reject the request, view the leave balance of employees.
- **Restrictions:** Cannot modify leave policies, cannot view leave requests outside their team.

Admin

- **Capabilities:** Manage user roles, view all leave requests, and generate leave reports.
- **Typical workflows:** Assign roles to users, approve or reject leave requests, generate leave reports.
- **Restrictions:** Cannot modify the core application settings, cannot delete user accounts.

Core Workflows

Submit a Leave Request

This workflow allows employees to request time off.

Steps:

1. Navigate to the 'Leave Requests' section by clicking on the 'Leave' tab in the main menu.
2. Click on 'Submit Leave Request' to open the request form.
3. Enter the required fields with values like:

- employeeId: 12345
- leaveType: SICK
- startDate: 2023-10-01
- endDate: 2023-10-05

4. Click 'Submit' to send the request.

5. **Expected outcome:** A confirmation message with the leave request ID and status as 'PENDING'.

Common mistakes:

- **Invalid employee ID:** Ensure the employee ID is correct and exists in the system.
- **Incorrect date format:** Use the format YYYY-MM-DD for dates.

Approve or Reject a Leave Request

This workflow allows managers to review and act on leave requests.

Steps:

1. Navigate to the 'Leave Requests' section by clicking on the 'Leave' tab in the main menu.
2. Click on the leave request ID to open the details.
3. Click 'Approve' or 'Reject' to change the status.
4. **Expected outcome:** The leave request status should update to 'APPROVED' or 'REJECTED'.

Tips:

- **Check leave balance:** Ensure the employee has sufficient leave balance before approving.
- **Provide feedback:** Use the comment section to provide feedback on the leave request.

View Leave Balance

This workflow allows employees to check their remaining leave balance.

Steps:

1. Navigate to the 'Leave Balance' section by clicking on the 'Leave' tab in the main menu.
2. Click on 'View Balance' to see the leave balance for different leave types.
3. **Expected outcome:** A list of leave types and their respective balances.

Data Management

Creating Records

To create a new leave request, navigate to the 'Leave Requests' section, click on 'Submit Leave Request', and fill in the required fields. Ensure all fields are correctly filled and validated before submission.

Editing Records

Existing leave requests can be modified by the employee who submitted the request. Navigate to the 'Leave Requests' section, click on the request ID, and then click 'Edit' to make changes. Managers can only change the status of a leave request.

Deleting Records

Leave requests cannot be deleted once submitted. However, they can be marked as 'CANCELLED' if necessary. This action is irreversible and should be used with caution.

Filtering and Search

Use the search bar and filters in the 'Leave Requests' section to find specific leave requests. Filters can be applied based on employee ID